

Reparametrization

Def. Let $\alpha: I \rightarrow \mathbb{R}^3$ be a Curve defined on an open interval I .

If $h: J \rightarrow I$ is a differentiable function on an open interval J , then the Composite $\beta = \alpha \circ h: J \rightarrow \mathbb{R}^3$ is a curve called a reparametrization of α by h .

Lemma. If β is the reparametrization of α by h , then the velocity is

$$\beta'(s) = \frac{dh}{ds}(s) \cdot \alpha'(h(s))$$

Proof. Denote $\alpha = (\alpha_1, \alpha_2, \alpha_3)$. Then, by the definition,

$$\beta(s) = (\alpha \circ h)(s) = (\alpha_1(h(s)), \alpha_2(h(s)), \alpha_3(h(s)))$$

Now, by the definition of velocity and the chain rule,

$$\beta'(s) = (\alpha_1'(h(s)) \cdot h'(s), \alpha_2'(h(s)) \cdot h'(s), \alpha_3'(h(s)) \cdot h'(s))$$

$$= h(s) \cdot (\alpha_1'(h(s)), \alpha_2'(h(s)), \alpha_3'(h(s)))$$

$$= h(s) \cdot \alpha'(h(s)).$$

Thm. Given a regular Curve α in $\mathbb{R}^3$,

there exists a reparametrization β of α such that $\|\beta'(s)\| = 1$ for all s .

Proof. Let $\alpha: I \rightarrow \mathbb{R}^3$ be a regular Curve. Let $a \in I$ be fixed.

Define $S(t) = \int_a^t \|\alpha'(u)\| du$. Since $\|\alpha'\| \neq 0$, the $S(t)$ is strictly increasing on I .

Therefore, there is an inverse function $t: J \rightarrow I$ s.t. $S(t(x)) = x$ for all $x \in J$, where $J = S[I]$. Now, for $\beta = \alpha \circ t$,

$$\beta'(s) = \frac{dt}{ds}(s) \cdot \alpha'(t(s)) \quad \text{and} \quad \|\beta'(s)\| = \underbrace{\frac{dt}{ds}(s)}_{>0} \|\alpha'(t(s))\| = \frac{dt}{ds}(s) \cdot \frac{ds}{dt}(t(s)) = 1. \quad \square$$

Note: given a curve α , define length $S(t) = \int_a^t \|\alpha'(u)\| du$ and $\beta = \alpha \circ S^{-1}$.

Def. Let $\beta: I \rightarrow \mathbb{R}^3$ be a unit-speed curve.

A vector field β' is called the unit tangent vector field on β , denote $T = \beta'$.

The derivative $T' = \beta''$ is called the curvature vector field of β .

The value $k(s) = \|T'(s)\|$ ($s \in I$) is called the curvature function of β .

To restriction s.t. $k > 0$,
the unit-vector field $N = \frac{T'}{k}$ is called the principal normal vector field of β .

The cross product $B = T \times N$ on β is called the binormal vector field of β .

Lemma. Let β be a unit-speed curve in $\mathbb{R}^3$ with $k > 0$,

Then, the three vector fields T, N , and B on β are unit vector fields that are mutually orthogonal at each point.

Proof. By definition, $\|T\| = \|\beta'\| = 1$. And since the curvature $k > 0$, $\|N\| = \frac{1}{k} \cdot \|T'\| = 1$.

And, since $\|T\|^2 = T \cdot T$ is constant, $2T \cdot T' = 0$, so $T \perp N$.

therefore the binormal, $\|B\| = \|T \times N\|^2 = \|T\|^2 \cdot \|N\|^2 - (T \cdot N)^2 = 1 - 0 = 1$.

In summary, $T = \beta'$, $N = T'/k$, $B = T \times N$, and $\{T, N, B\}$ is orthonormal.

Since $B \cdot T = 0$, $B' \cdot T + B \cdot T' = 0$ by differentiate. Therefore,

$$B' \cdot T = -B \cdot T' = -B \cdot kN = 0$$

And since B is unit, $B' \cdot B = 0$, thus, there is a scalar value function τ s.t.

$$B' = -\tau N$$

, called the torsion. Now

Thm. Let β be a unit-speed curve in $\mathbb{R}^3$ with curvature $k > 0$ and torsion τ ,

$$\begin{aligned} T' &= kN \\ N' &= -kT + \tau B \\ B' &= -\tau N \end{aligned} \quad \begin{bmatrix} 0 & k & 0 \\ -k & 0 & \tau \\ 0 & -\tau & 0 \end{bmatrix} \begin{bmatrix} T \\ N \\ B \end{bmatrix}$$

Proof. The first and third rows are just definition. To show the second, observe $N' = (N' \cdot T)T + (N' \cdot N)N + (N' \cdot B)B$

Now, $N' \cdot T = N \cdot T' = 0$ and $N' \cdot B = N \cdot B' = 0$ gives the results.

Prop. Let β be a unit-speed curve in $\mathbb{R}^3$ with $\kappa > 0$. Then,

β is a plane curve if and only if $\tau = 0$, the torsion is zero.

Proof. Suppose that β is a plane curve. That, $\text{Im } \beta$ lies in some plane.

Therefore, there are two vectors $p, q \in \mathbb{R}^3$ such that $q \neq 0$ and

$$[\beta(s) - p] \cdot q = 0 \text{ for all } s.$$

Differentiating yields:

$$\beta'(s) \cdot q = 0 \text{ and } \beta''(s) \cdot q = 0 \text{ for all } s.$$

That is, q is orthogonal to $T = \beta'$ and $N = \beta''/\kappa$. So, β and q are parallel.

We can conclude that $\beta = \pm q/||q||$. So $\beta' = 0$, the torsion τ must be zero.

Conversely, suppose that the torsion is zero. Then, $\beta' = 0$ by definition.

($\beta' = 0$ means β is parallel.) To show that $\text{Im } \beta$ lies in some plane, define

$$f(s) = [\beta(s) - \beta(0)] \cdot \beta(s) \text{ for all } s.$$

Then, the derivative is

$$f'(s) = \beta'(s) \cdot \beta(s) + \beta(s) \cdot \beta'(s) \stackrel{\text{def.}}{=} \beta'(s) \cdot \beta(s) = T \cdot \beta = 0.$$

By definition, $f(0) = 0$, so $f = 0$. Consequently,

$$[\beta(s) - \beta(0)] \cdot \beta = 0 \text{ for all } s,$$

So proved $\square$

Prop. Let β be a unit speed curve with constant curvature $\kappa > 0$ and torsion zero.

Then, the β is part of a circle of radius $1/\kappa$.

Proof. Since $\tau = 0$, the β is a plane curve. Define a curve

$$\gamma(s) = \beta(s) + \frac{1}{\kappa} \cdot N(s).$$

By the Frenet formula, $\gamma' = \beta' + \frac{1}{\kappa} \cdot N' = T + \frac{1}{\kappa} \cdot (-\kappa T) = 0$. Therefore, γ is constant.

Now, $d(\gamma, \beta(s)) = \|\gamma - \beta(s)\| = \frac{1}{\kappa} \cdot \|N\| = \frac{1}{\kappa}$, so proved.

General version of Frenet formula,

Let $\alpha: I \rightarrow \mathbb{R}^3$ be any regular curve, not necessary $|\alpha'| \equiv 1$.

And, let $\beta: J \rightarrow \mathbb{R}^3$ be a reparametrization of $\alpha[I]$ by the arc-length

$$s = s(t) = \int_{t_0}^t |\alpha'(u)| du, \quad \text{where } t_0 \in I.$$

with the inverse function $t = t(s) = s^{-1}$. Then,

$$1). \frac{dt}{ds} = \frac{1}{|\alpha'|} \quad \text{and} \quad \frac{d^2t}{ds^2} = - \left(\frac{\alpha' \cdot \alpha''}{|\alpha'|^4} \right)$$

2). The curvature and torsion of α is given by

$$K(t) = \frac{|\alpha' \times \alpha''|}{|\alpha'|^3} \quad \text{and} \quad \tau(t) = - \frac{(\alpha' \times \alpha'') \cdot \alpha''}{|\alpha' \times \alpha''|^2}.$$

Signed Curvature for plane curve.

Let $\alpha: I \rightarrow \mathbb{R}^2$ be a plane curve.

Then, the binormal B is constant vector, so we consider only T and N .